



Artificial Intelligence Enhanced Digital Learning for the Sustainability of Education Management System

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ABSTRACT

Maintenance schedules are scheduled ahead of time and automatically based on the continuous monitoring of the equipment by statistical methods, thanks to artificial intelligence-enabled digital transformation and the best fit model based on Machine Management Index in a pedagogical system. One of the most important aspects of universities is the widespread use of machine learning methods to evaluate students' progress. Machine learning approaches are designed to speed up the learning process without sacrificing accuracy. The dynamics of teaching and learning have shifted since the introduction of modern technological tools. The educational system as a whole has changed and developed over time. These days, people can get an education outside of the classroom as well, thanks to the proliferation of online courses and resources. Everyone's professional life begins with their education. By analyzing past data, artificial intelligence methods can resolve existing problems. When applied properly, artificial intelligence can be a highly efficient method for solving problems with a predictable and repeatable solution space. The learner's personality can be predicted based on a number of factors using machine learning approaches. This article examines how AI may improve digital learning in education management systems to sustain the education ecosystem. AI in education improves student results, learning experiences, and administrative processes. This study discusses AI applications in education management systems and associated problems and opportunities. We also explore ethical issues and the roadmap for using AI to improve education. Educational institutions can provide individualized curriculum for students based on their unique personalities and areas of interest. Institutions of higher learning can benefit greatly from this instrument for personality prediction by recommending a course of study that will better prepare students to enter the field of their choice and achieve professional success.

1. Introduction

The original goal of AI was to develop computer programs with capabilities similar to human intelligence. The methods of artificial

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intelligence re-create the thinking process and its result in an even more modest fashion (Dai et al., 2020). The role of AI approaches in educational content is not to replicate human knowledge exactly from the outset of the field. These methods are used to facilitate better student-teacher relationships. If the system can prove that it can do what it claims it can do, it will be able to have a conversation about the problem, provide compromise solutions, and explain why it made the decisions it did based on the student's input. The shift in perspective from "student as thinker" to "reasoner with student" Human development studies that take insight not as the result of fixed knowledge structures but as the capacity to relate to our social and physical condition will find this development consistent (Knox, 2020a). A network of researchers interested in the integration of AI into educational settings is a strong indicator that these areas are converging. Artificial intelligence's likely presentation of attention-taking capacity is a major contribution to the advancement of educational innovation. Education provides the context for learning the framework. Artificial intelligence (AI) strategies are motivated by a desire to build the right arrangement quickly and efficiently. We already know that there is a critical shortage of teachers in India. Furthermore, there has been no comparable improvement in the quality of educational strategy or pedagogy. Our students have been the subject of repeated press reports suggesting that they are being denied access to a high-quality education (Knox, 2020b). Artificial intelligence simulations could be the solution to this problem. Teachers have many responsibilities, including testing, grading, paper setting, student evaluations, and monitoring classroom performance. If these tasks were simplified for teachers, they might devote more time to enhancing their courses and their students' learning and performance (Boulet & Durning, 2019). The primary justification is the adaptability of artificial intelligence frameworks to the unique learning and controlling abilities of each student.

Also, it might unearth slimmer attributes and flaws. When there are sixty students in a study hall, one teacher isn't enough to meet their needs. To solve the dual problems of poor quality and scarcity, it may be best to equip students with AI frameworks in classrooms, study spaces, and at home (Cukurova et al., 2019; Du Boulay, 2019; Kay & Kummerfeld, 2019; Li et al., 2018; Li et al., 2022; Ruby et al., 2022). Teachers can be assisted in all of these tasks by computer-based intelligence frameworks, which makes them both mechanical and intellectual in nature. With AI frameworks in place, teachers will have more time to work directly with students instead of on mundane administrative tasks. Using Artificial Intelligence, we can expand the range of intellectual tasks we're capable of as humans. Artificial intelligence may investigate an activity's root cause and significance, as well as employ novel approaches to learning from experience in order to choose the best course of action to take (Aslam et al., 2021). Thanks to AI, computers may now function as autonomous agents, capable of learning and thinking. An artificially intelligent agent is able to learn new things through data collection, reason with the use of rules to produce predictable outcomes, and adjust its behavior on the fly. Students believe that AI can help schools automate their grading processes. The AI system can tailor its responses to each individual student, allowing them to progress through the material at their own pace (Murtaza et al., 2022a). Artificially intelligent instructors can provide further aid to students, putting them on the right path. Recruiters can use AI to take a more proactive and strategic approach to hiring. Investing more time in applicants allows you to get to know them and assess their cultural fit.

AI has made learning personalized and interactive. AI systems evaluate massive quantities of data to personalize learning routes to each student's strengths and shortcomings. Adaptive learning platforms alter content based on student achievement to maximize challenge. Intelligent Tutoring Systems provide tailored guidance and fast response like human tutors. NLP also helps language learners with interactive language practice and pronunciation feedback. Gamification and interactive aspects make learning fun, engaging pupils. Data-driven insights and learning analytics help educators optimize training for better learning outcomes. AI-powered personal assistant bots free teachers to teach and guide. AI-enabled continuous learning platforms also curate customized courses for lifelong learning and upskilling. AI in education can improve efficiency, engagement, and equity. AI affects schooling beyond its technological capabilities. AI improves tailored learning experiences, motivating and retaining students. AI's capacity to process large information helps educators understand student performance and provide customized guidance. Gamified learning and interactive components make learning fun and engaging. AI technologies also streamline administrative work, giving educators more time to focus on personalized teaching and mentoring, establishing meaningful student-teacher interactions. To enable responsible and equitable AI integration in education, ethics and data privacy must be addressed as AI advances. AI is enabling continuous and tailored learning. Students can learn at their own pace and avoid frustration using AI-powered adaptive learning solutions. Instant, focused feedback from Intelligent Tutoring Systems boosts students' confidence and learning. NLP simulates real-life language use and provides pronunciation feedback, improving language fluency. AI-driven data analytics also help instructors make evidence-based decisions to improve instruction. Responsible AI adoption may create a future-ready learning ecosystem that prepares students for a changing world.

2. Background

Many aspects of digital education could benefit greatly from the application of artificial intelligence (AI) (Murtaza et al., 2022b). The manuscript uses AI to improve school management and learning. Intelligent tutoring systems, adaptive learning platforms, natural language processing, gamification, data analytics, and personal assistant bots are covered. AI technologies improve data-driven decision-making, personalized learning, and administrative efficiency. The book emphasizes how AI may increase student results, engagement, and retention by customizing instructional content, delivering real-time feedback, and creating a more immersive learning environment. The following are some of the ways in which AI can improve online education:

2.1.1. Personalization

AI may assess information about students' preferences, learning methods, and performance to create individualized lessons. Having access to the materials that are most pertinent to their requirements may keep students interested and motivated. Personalized learning for each individual student is a major benefit of employing AI in digital education. Artificial intelligence (AI) algorithms can personalize education by assessing information about students' learning preferences, interests, and progress. As a result, pupils are more likely to remain interested and motivated, which in turn leads to better learning.

2.1.2. Adaptive learning

With the use of artificial intelligence, adaptive learning systems may modify the level of challenge for students based on their performance. Learning at one's own pace like this can assist students avoid becoming stuck on content that is too simple or too challenging for them. Adaptive learning experiences that are tailored to each student's current level of understanding are another major upside of employing AI in online education (Mohanta & Mandal, n.d.; Bin & Mandal, 2019; Davis, 2019; Donkin et al., 2019; Ouyang et al., 2022). Data on student performance and progress can be analyzed by adaptive learning algorithms, which then modify the course's difficulty level. This can make sure that children are learning at a pace that is right for them and keep them from getting bored or frustrated.

2.1.3. Intelligent tutoring systems

Artificial intelligence (AI) has the potential to develop tutoring systems that are as helpful to students as human teachers. Learners can get instantaneous feedback from these systems, which can aid in the identification of problem areas and the provision of individualized remediation. Tutoring systems driven by AI can provide each student specific advice based on their own strengths and weaknesses. Technology like natural language processing allows these systems to interpret student queries and deliver personalized responses and grading. This can ensure that students receive the guidance and assistance they need to reach their full potential.

2.1.4. Natural language processing

Natural language processing driven by AI enables students to converse with chatbots and voice assistants in a more natural way with digital content. For certain students, especially those who have difficulty concentrating on printed language, this can make the learning process much simpler and more natural. Students can benefit from chatbots and other voice assistants that use natural language processing powered by AI. This can help students who have difficulty reading or writing get the answers they need faster and make learning more accessible and intuitive for all students.

2.1.5. Predictive analytics

Artificial intelligence can evaluate student data to determine which students are at risk of failing a course or having difficulty with a particular topic. This can help teachers catch these students before they fall too far behind and give them extra aid. Finally, AI can be used to foresee which students could struggle with a given topic or lag behind their peers. Data on student achievement and conduct can be analyzed by predictive analytics algorithms to help teachers identify children who could benefit from extra help. In the long run, this can boost student achievement because it allows teachers to catch problems early on and correct them.

To sum up, AI is becoming increasingly instrumental in the improvement of online education. AI technology can help students thrive and reach their full potential by creating learning environments that are unique to each individual (Kneale, 2018). There must be an emphasis on ethics, transparency, and student preferences when implementing AI in the classroom. In general, AI has the potential to significantly improve the quality, efficiency, and applicability of online education. It is essential, however, that AI be used in a way that is ethical, transparent, and tailored to the requirements and preferences of the students.

AI concepts can transform rural schooling. AI can help rural schools provide superior education. AI-powered adaptive learning solutions can customize instructional content for rural pupils even in resource-constrained contexts. Intelligent tutoring technologies can provide tailored support and direction to students without enough human instructors. AI-driven data analytics can identify learning gaps and track student progress, enabling tailored interventions and support for challenging learners. Natural language processing helps AI create tailored and culturally relevant instructional content. Local language learning tools make rural education more accessible and interesting. Gamification and interactive learning can motivate rural pupils by making education fun and engaging. To use AI in rural settings, various problems must be addressed. Internet connectivity and infrastructure may hinder AI-powered platform integration. AI-driven products should be made offline so remote students can access educational content. Responsible use of AI in rural education should protect students' personal data and ensure that AI algorithms do not perpetuate biases or discrimination. In conclusion, AI concepts improve education in rural places. Educational institutions can improve learning results, tailor support, and meet student needs with AI. AI can provide quality education and localize content in rural places. To empower rural education with AI, thorough infrastructure, ethical, and privacy planning is needed.

3. Methodology

New strategies for early identification of struggling pupils are needed if we are to raise academic achievement overall. The number of mishaps increases when attention is delayed. Therefore, a potential cutting-edge technological option for earlier identification of

student performance is the merging of machine learning and information technology. Feature selection is a crucial but time-consuming part of the model-building process (Ray, 2017). Here, we select the variables that will be used to make forecasts. Features can be prioritized in order of significance and then selected accordingly. Many machine learning algorithms are given in the literature for properly forecasting student performance. Some examples of these algorithms are the Nave Bayesian technique, logistic regression, multi-layer perceptron, neural network, and others. Risk factors for student performance can be determined with the use of a few machine learning algorithms that use feature selection techniques. It is also noted that students are utilizing a variety of ensemble techniques in their presentations. We design and build a custom online learning platform with four distinct sections (Demartini et al., 2020). The goals of this model are to analyze the learner's behavior, determine their preferred learning style, anticipate their domain knowledge, and provide recommendations for e-learning content and a personalized learning route. The proposed architecture for an AI-based individualized online learning environment is depicted in Fig. 4. (See Figs. 1–3.) (See Figs. 5–8.)

The goal of artificial intelligence is to give a computer the ability to reason and problem-solve in much the same way as a human being would. Artificial intelligence, or AI, is a subfield of computer science focused on creating machines that mimic human intelligence in terms of both reasoning and action. It operates on the backs of many artificial intelligence and data science methods such as machine learning, deep learning, heuristic searching, rule-based systems, ANNs, SVMs, NLP, etc. Dual-intelligence systems exist. Artificial intelligence techniques that are narrow or weak simulate human intelligence but only in specific contexts. Narrow AI is AI that just cares about getting one specific job done well. These machines are highly intelligent systems, capable of working within a wide variety of limitations and restrictions, often outperforming even the most fundamental forms of human intelligence. The form of

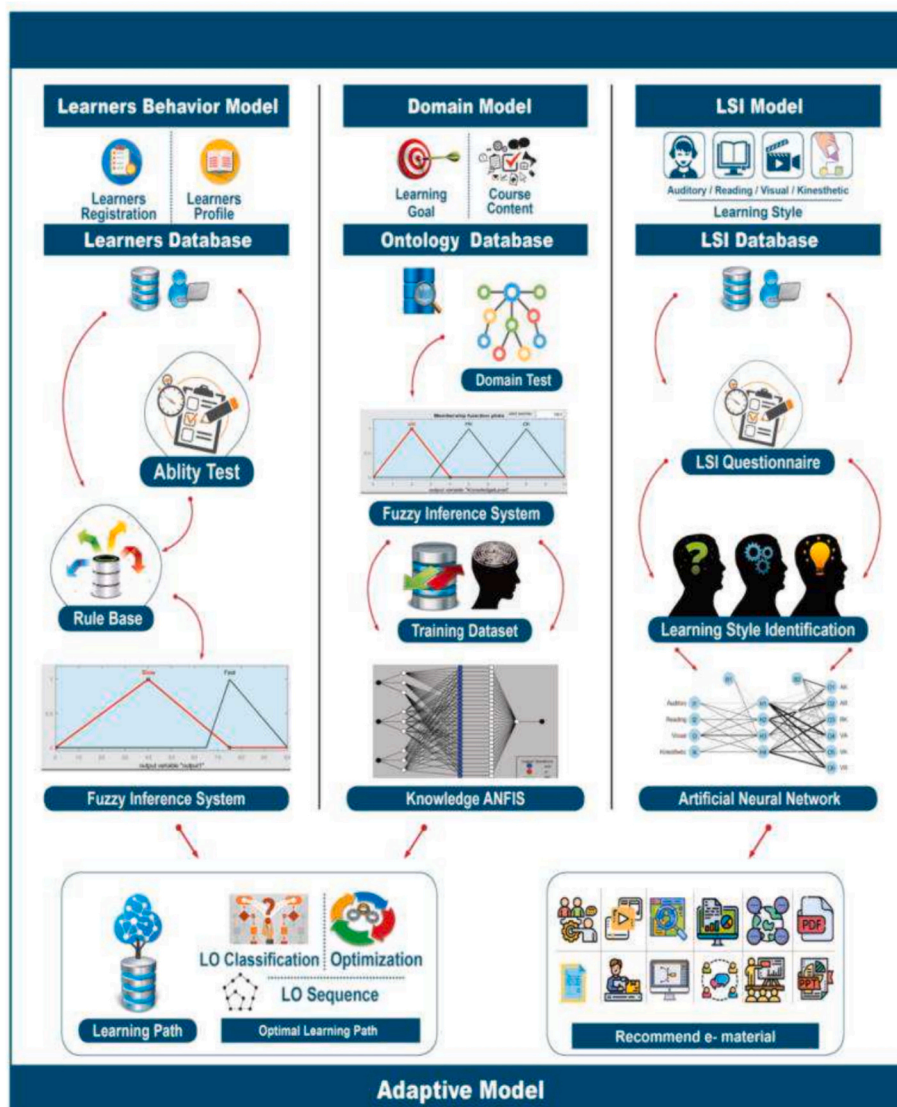


Fig. 4. Adaptive model for AI enabled Digital learning.

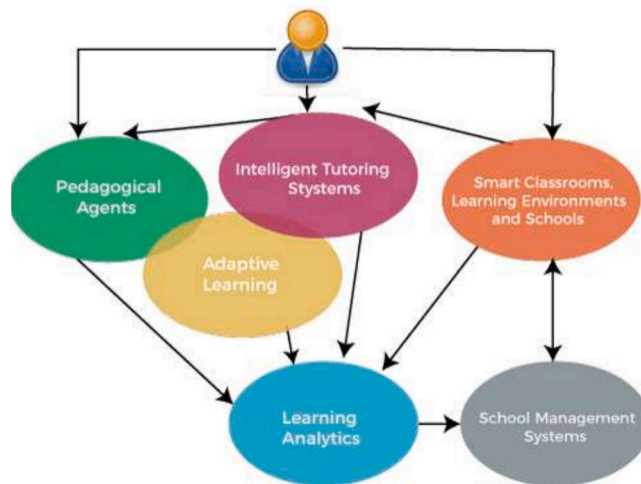


Fig. 1. AI based Enhanced Digital Learning.

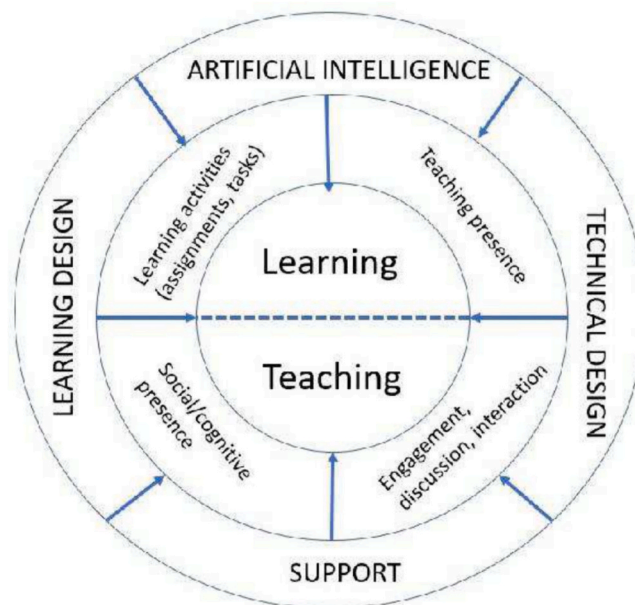


Fig. 2. Learning Teaching process using AI.

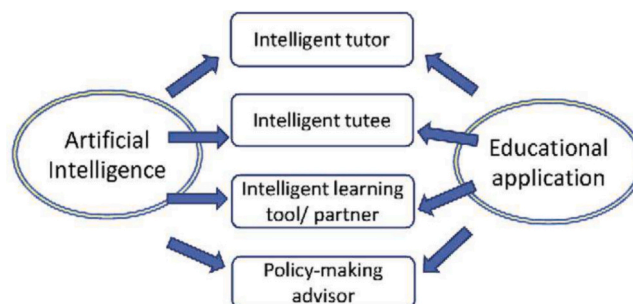


Fig. 3. Role Artificial Intelligence in enhancement of Digital Learning.

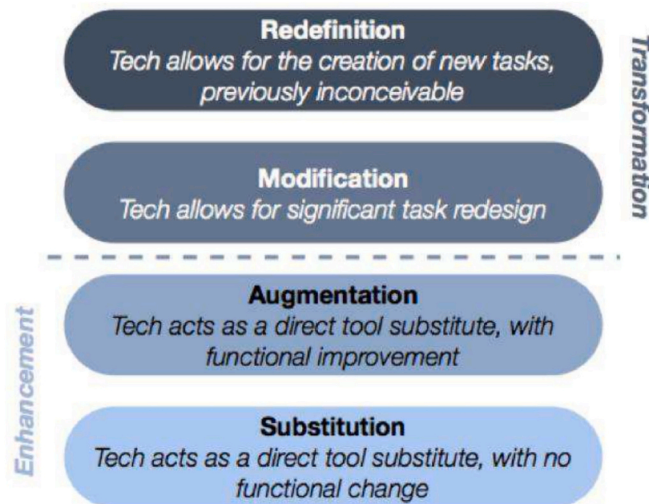


Fig. 5. AI with enhanced Digital Learning Model using SAMR.

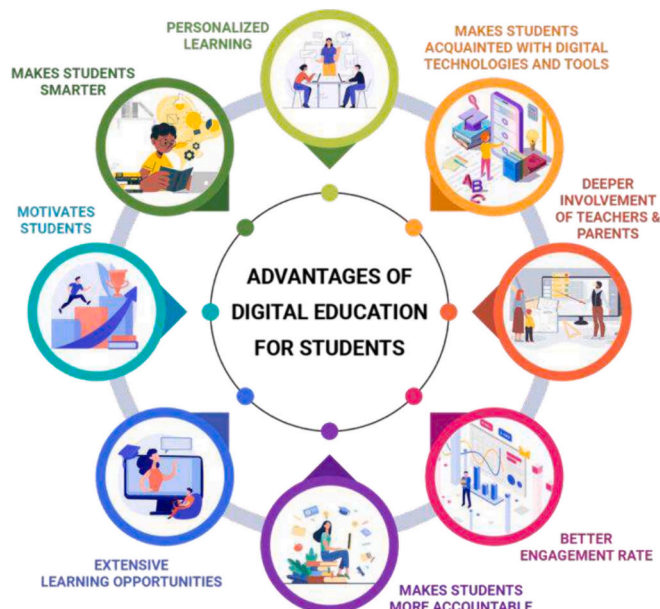


Fig. 6. Advantages of Digital education of the students.

artificial intelligence seen in robots, movies, etc. is known as strong AI or Artificial General Intelligence (AGI). It is an intelligent machine that can reason and do tasks like a human being; it can use this intelligence to find answers to any difficulty. Smart assistant systems, such as Alexa, Siri, and Cortana, are just one example of the many applications of artificial intelligence. These smart voice recognition systems not only help us learn more, but also make better decisions.

Artificial intelligence (AI) plays a crucial part in the automation of numerous operations across all industries, from manufacturing and medicine to e-commerce and education. By catering to each individual student, artificial intelligence methods are revolutionizing education. The true potential of AI is in its ability to retain and analyze massive amounts of data on students in order to tailor instruction to each learner. If a teacher doesn't have enough background knowledge in a certain topic to properly assess each student's strengths and interests, AI can step in to help. There is a branch of machine learning called deep learning. Machine learning algorithms aid in the analysis of brain function by way of an artificial neural network. Deep learning is a clever method for processing large data sets efficiently. It's used to create AI systems that can learn input-output mapping of complex functions from raw data, bypassing the need for hand-crafted human features. By using AI-based methods, such as Fuzzy rule-based systems and artificial neural networks, a customized e-learning system can be created.

An approach to understanding the potential effects of technology on education is provided by Dr. Ruben R. Puentedura's 2006

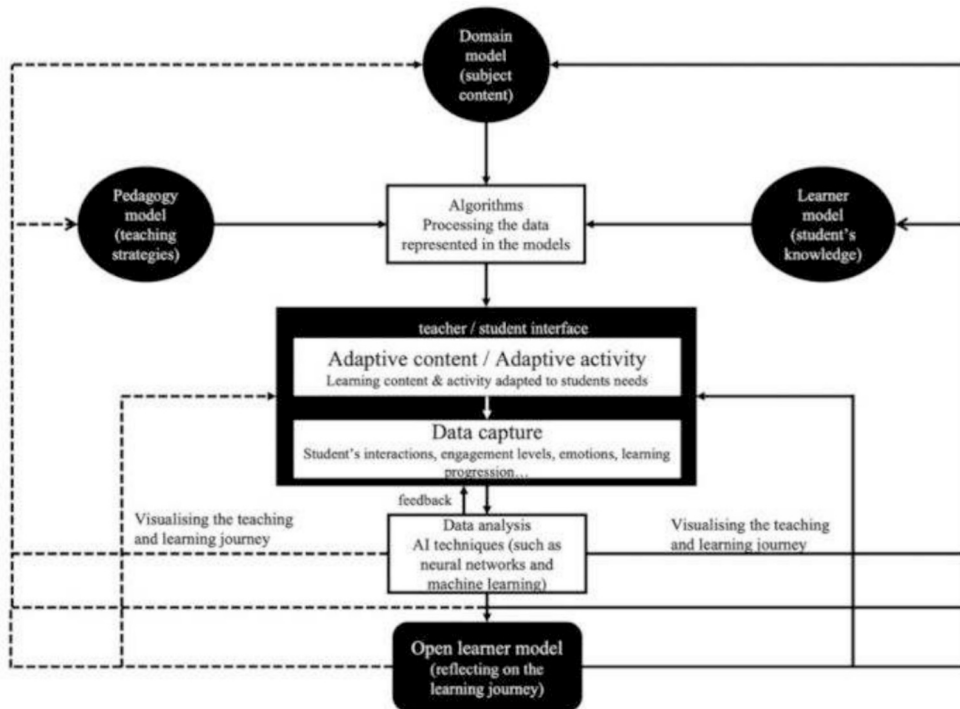


Fig. 7. Process of AI based Data Analysis of Enhanced digital Learning.

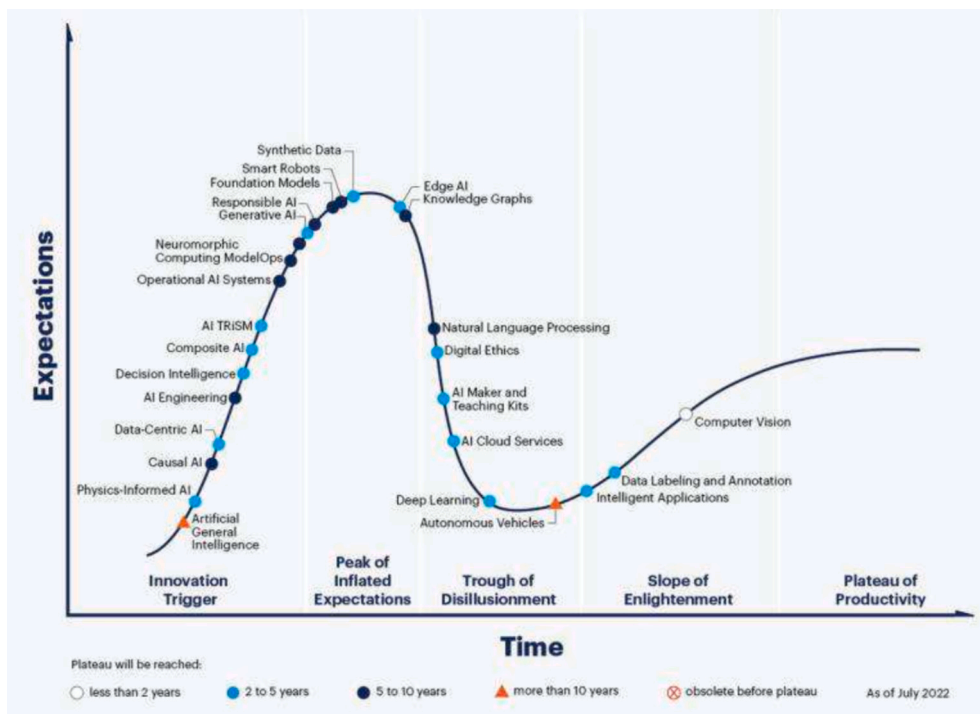


Fig. 8. Hype cycle for artificial intelligence in enhanced digital learning.

Substitution Augmentation Modification Redefinition (SAMR) Model. It also illustrates a common path taken by early adopters of edtech as they integrate it into their own pedagogy and classrooms. In order to tell the difference between human and machine intelligence, AI, or Machine Intelligence (MI) as Seldon and Abidoye called it, is defined as a human-centered machine's ability to

understand its surroundings and make adjustments to maximize its chances of success. “Mechanically calculate logical statements for achieving objectives” defines artificial intelligence in this way. The aphorism emphasizes the machine’s ability to think and adapt intelligently using a “logical and linear structure,” implying that intelligence is the linear computation of data-driven facts, which raises questions about artificial intelligence’s philosophical foundations. Replace AI/machine intelligence with augmented intelligence.

4. Discussion

The adaptive paradigm for AI-enabled digital learning can change education by giving students personalized learning experiences. The adaptive approach uses AI to analyze each student’s learning patterns, preferences, strengths, and limitations to create personalized learning routes. Personalized learning challenges pupils without overwhelming or boring them. The adaptive model boosts engagement, motivation, and subject mastery, improving learning outcomes. Its flexibility and inclusivity allow students of all learning styles and abilities, particularly those in rural or underserved places, to succeed. The adaptive model’s data-driven nature allows for continual improvement and evolution, enabling educators to alter teaching tactics and content based on useful insights. The adaptive model caters to learners of all ages and backgrounds, encouraging lifelong learning and skill development. This article analyzes AIED’s methodology, programs, and resources. This paper answers “What do we mean by Artificial Intelligence in Education?” It considers important implications of AIED, such as the ethical concerns and digital competencies teachers would need to develop to embrace and transform discourse and rethink their roles as educators, positioning intelligent computational representations as sophisticated scaffolds that may help students improve their learning experience and intellectual abilities. This merger of traditional teaching approaches with artificial intelligence as a complement could revitalize education planning, implementation, and evaluation. Educators who want to answer “What would it mean to teach and learn in the age of AI?” must assess the positives and cons of using AI. Seldon and Abidoye term education the “Cinderella of the AI story” (Dai et al., 2020) since AI in education is still in its infancy and receives little attention. Holmes et al. (Knox, 2020a) believed it was wishful thinking to imagine that AI wouldn’t impact education in some way, whether technological, pedagogical, ethical, or connected to teacher training.

With the rise of AI technology, it’s more important than ever to stay abreast of developments in your field. If you want your pupils to benefit as much as possible from your teaching, you need to become an expert in your field. Participating in professional development opportunities, going to conferences, making connections with colleagues, and reading the latest research and publications are all great ways to achieve just that. If you do this, you’ll be able to give your students new and interesting points of view. Then, keep in mind that AI provides novel ways to motivate students and develop an individualized curriculum. You can experiment with innovative methods of instruction that make use of AI-enhanced tools to analyze and respond to each student’s unique learning requirements. We pick up knowledge and skills from our peers and via our interactions with them. Parts of the learning process can be replaced by artificial intelligence, but social connections are still necessary. You’ll stand out from the competition if you provide students with access to a supportive learning environment and knowledgeable teachers.

Tools and platforms powered by AI can help you save time in the classroom by automating routine administrative activities and giving you deeper insights into students’ progress. For instance, a learning management system can monitor student development and pinpoint trouble spots. Natural language processing methods can also be used to evaluate student work and offer suggestions for improvement in areas such as grammar and style. You may make your job as a teacher easier and more efficient while also enhancing your students’ learning experiences by using AI-powered tools and platforms.

By repeatedly extracting and distinguishing domain, pedagogy, learner, and open learner models, AI algorithms might infer adaptive material and individualized learning activities. The ITS system may learn from this cycle. Experience-based learning connects students to education. The ITS connects the student to a specific educational experience by specifying the “what” (domain model), “how” (pedagogy model), and “who” (learner model). This relational approach to processing computational models is still in its early stages, but it could help move away from a deterministic view of AIED as a replacement for traditional pedagogical practices and toward a more relational approach that integrates the student’s experience. To achieve complete relationality, the domain, pedagogy, and learner models must be decomposed to lower sub-model levels to provide more meaningful and integrated relationships.

Both positive and negative results are magnified when AI is used in place of human judgment. The proper results can be achieved with the help of responsible AI since it solves problems associated with providing value vs accepting risks. Business and social benefit, risk, transparency, fairness, bias reduction, explainability, responsibility, safety, privacy, and regulatory compliance are all components of responsible AI, which encompasses all of these and more. It will take 5–10 years for responsible AI to reach general acceptance, but its influence on business will be revolutionary. Within the next two to five years, digital ethics will likely become a major corporate trend. The term “digital ethics” refers to a body of values and principles for the proper use of technology in relationships between individuals, groups, and entities. Many people are still worried about these problems, especially when it comes to privacy and prejudice. People are becoming more and more annoyed by the lack of openness, abuse, and breaches involving their personal information. Companies are taking precautions to ensure the privacy and safety of customer information as governments enact stronger regulations.

5. Conclusion

Artificial intelligence (AI) is influencing the future of education. This presents both fascinating new prospects and problems for the field of education. Teachers of the future will need to strike a balance between relying on AI and engaging their students in meaningful conversation. Teachers need to pay attention to their areas of expertise, strengthen their learning communities, and design exciting lessons for their pupils. “What do we mean by Artificial Intelligence in Education?” extensive literature review. The authors used

thematic analysis and synthesis to position the study in the context of educational technology research. AIED's adaptive and personalized nature helps students learn and build skills that are most relevant to them. The use of artificial intelligence (AI) is crucial to improving online education. Personalization, adaptability, and interactivity are all possible in digital learning thanks to AI. In general, AI has the potential to greatly improve digital learning by making it more individual, flexible, and engaging. In addition to assisting students in meeting their goals, AI-based solutions can also shed light on where and how teachers might make improvements. In conclusion, integrating AI into education administration systems has the potential to improve the overall education ecosystem. AI-enhanced digital learning improves education quality, efficiency, and accessibility. AI technology like intelligent tutoring systems, predictive analytics, and adaptive learning platforms can give students individualized and interesting learning experiences, improving academic performance and retention. However, integrating AI into education management entails addressing ethical issues, data protection problems, and human-AI balance. Fairness, transparency, and student equity must be prioritized in AI implementation. Ethical frameworks and rules require ongoing collaboration between educational institutions, AI specialists, researchers, and policymakers. AI-enhanced digital learning can help raise a generation of lifelong learners as technology advances. We can create a more inclusive and resilient education system that meets the requirements of students worldwide by using AI to adapt and personalize instruction. A sustainable, effective AI-powered education landscape requires vigilance, collaboration, and a dedication to high-quality education for all.

CRedit authorship contribution statement

K.S. Suryanarayana: Conceptualization, Methodology. **V.S. Prasad Kandi:** Software, Data curation. **G. Pavani:** Visualization, Investigation. **Akuthota Sankar Rao:** Writing – original draft. **Sandeep Rout:** Supervision. **T. Siva Rama Krishna:** Writing – review & editing.

Data availability

Data will be made available on request.

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